

METplus: Continuous integration with GitHub Actions for reproducible results and reduced software vulnerabilities

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METplus



Community Software Considerations

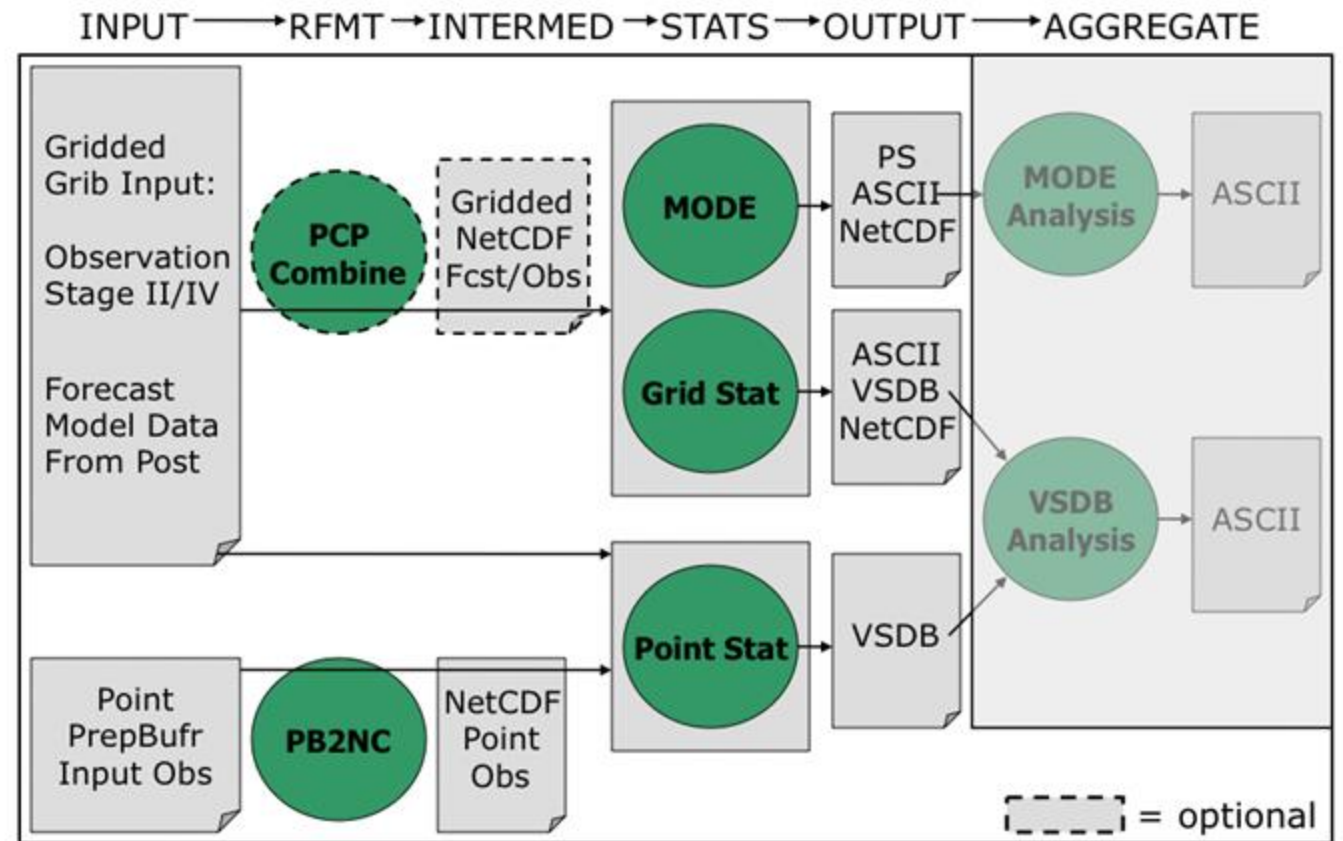
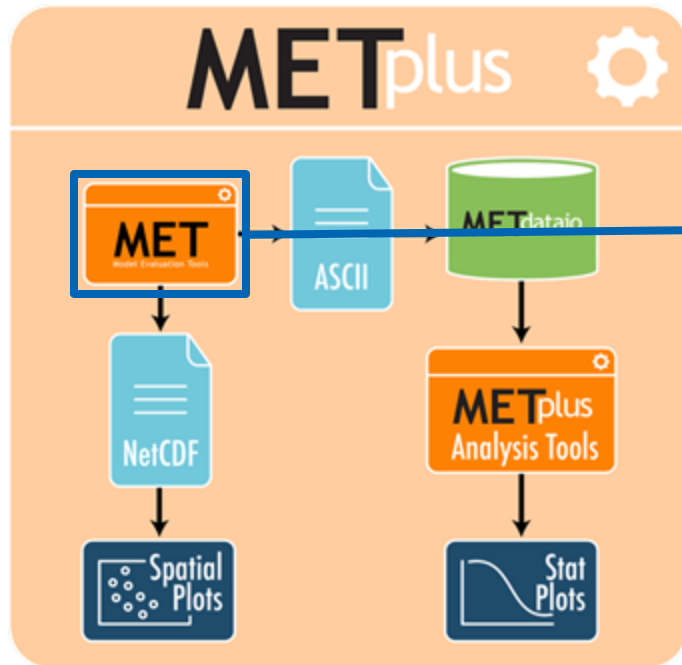
more than new features

- Funding for functionality and **New Features**
 1. Version control (cvs, svn, git)
 2. Continuous integration (regression testing and test coverage)
 3. Continuous deployment (containers, conda)
 4. Code quality (Fortify, SonarQube static code analysis and optimization)
 5. Security scanning (CVEs and STIGs)
 6. Development and release workflows (new features and bugfixes)
 7. Internal communication (email, meetings, Slack)
 8. Project management (Jira, GitHub issues, PRs, and projects)
 9. Documentation (Users, Contributors, and Developers, DOIs)
 10. User training (online tutorial, pre-recorded videos, live sessions)
 11. User support for sponsors and community (email, GitHub discussions)
 12. User outreach and marketing (proposals, presentations, social media)

Functionality through Time

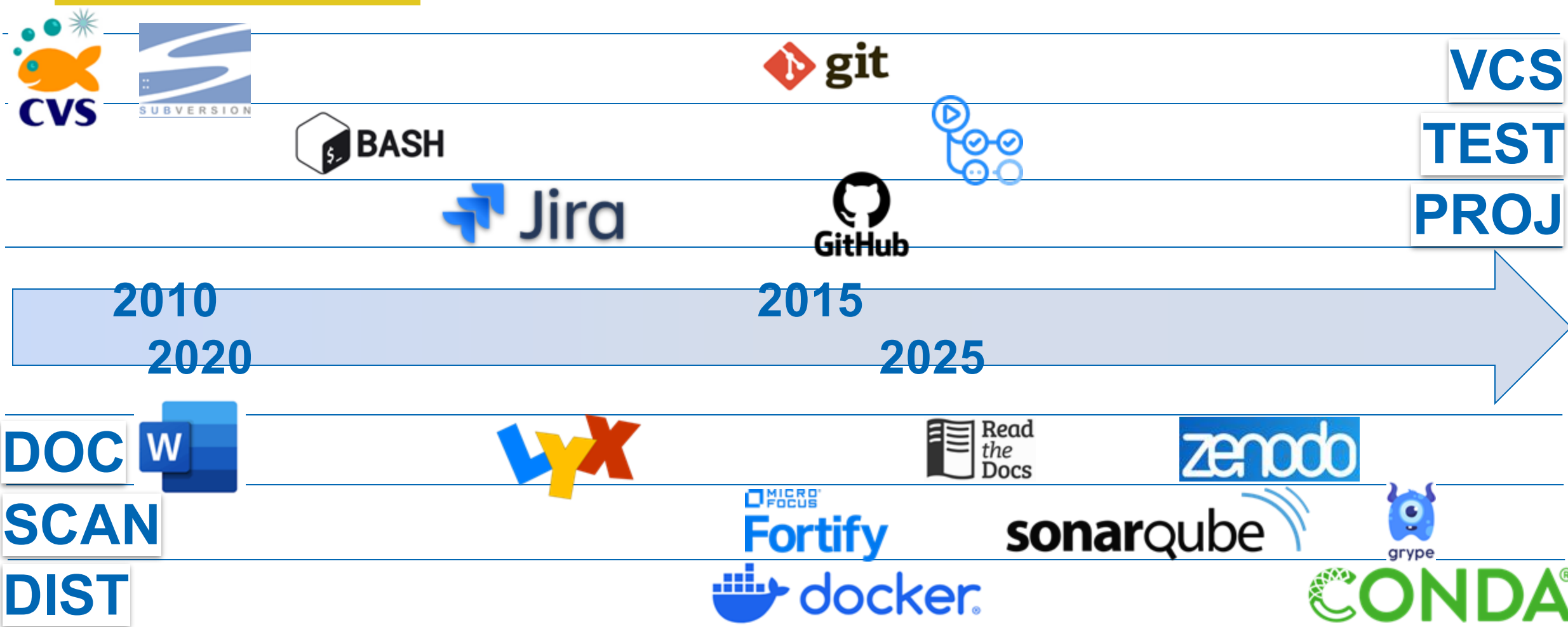
growth and expansion

MET Overview v0.9



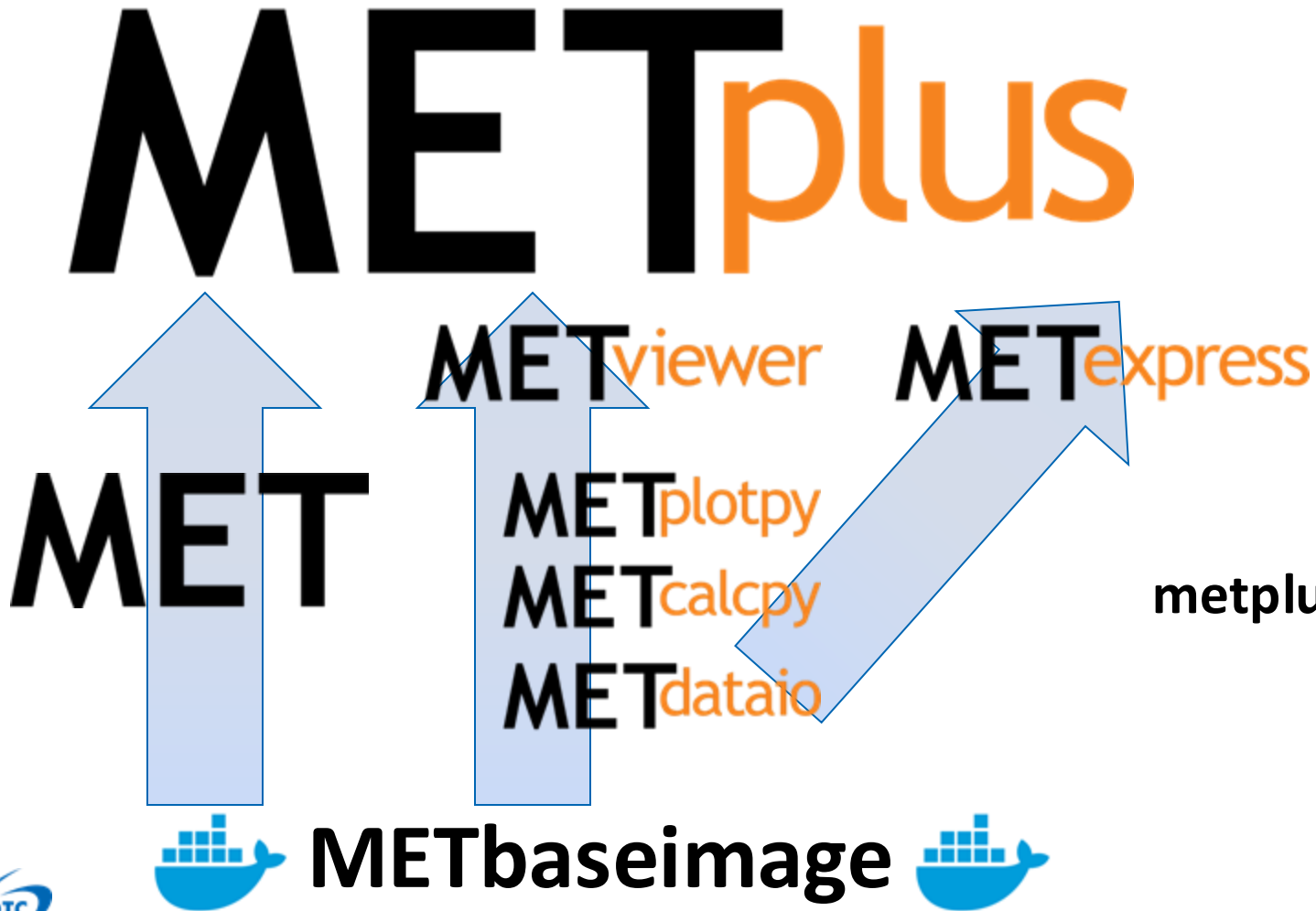
Support Tools through Time

evolution



DTCenter GitHub Organization

repository hierarchy



Nonprofit Status

GitHub Teams

- 77 users (\$3.7K)
- Actions (\$6K)



Docker Hub DSOS

- 10 users (\$2K)



Slack Pro

- 23 Users (\$2K)



metplus-action-...

scan-docker-images, free-disk-space,
data-update, trigger-use-case,
release-checksum, sonarqube,
build-documentation

METplus Release Guide (RTD)

metplus.rtf.d.io

Release Guide
Release Types
Release Support Policy
Release Dependencies
Instructions Summary
1. Coordinated Release
2. METplus Release
3. MET Release
4. METdataio Release
5. METcalcpy Release
6. METplotpy Release
7. METviewer Release
8. METexpress Release
9. Recreate an Existing Release

[Home](#) / [Release Guide](#) / [3. MET Release](#) / 3.1. MET Official Release [View page source](#)

3.1. MET Official Release

Create a new vX.Y.Z official release from the develop branch.

3.1.1. Open Release Issue

Define a new GitHub issue for creating this new release.

Instructions >

3.1.2. Clone the Project Repository

Clone the repository

Instructions

```
1 MET Official Release
2 -----
3
4 .. [projectRepo] replace:: MET
5
6 Create a new vX.Y.Z official release from the develop branch.
7
8 .. include:: release_steps/open_release_issue.rst
9 .. include:: release_steps/clone_project_repository.rst
10 .. include:: release_steps/checkout_main_branch.rst
11 .. include:: release_steps/create_release_feature_branch.rst
12 .. include:: release_steps/met/update_version_official.rst
```

Name
..
Contributors_Guide
Release_Guide
Users_Guide
Verification_Datasets

☒ Do these changes include sufficient documentation updates, ensuring that no errors or warnings exist in the build of the documentation? **[Yes]**

METplus

METplus

latest

Search docs

TRAINING

METplus Tutorial

Training Series

Featured Topics

METPLUS

User's Guide

Contributor's Guide

Verification Datasets Guide

Release Guide

MET

User's Guide

Contributor's Guide

METEXPRESS

User's Guide

METVIEWER

User's Guide

Contributor's Guide

METPLOTPTY

User's Guide

Contributor's Guide

METCALCPY

User's Guide

Contributor's Guide

METDATAIO

User's Guide

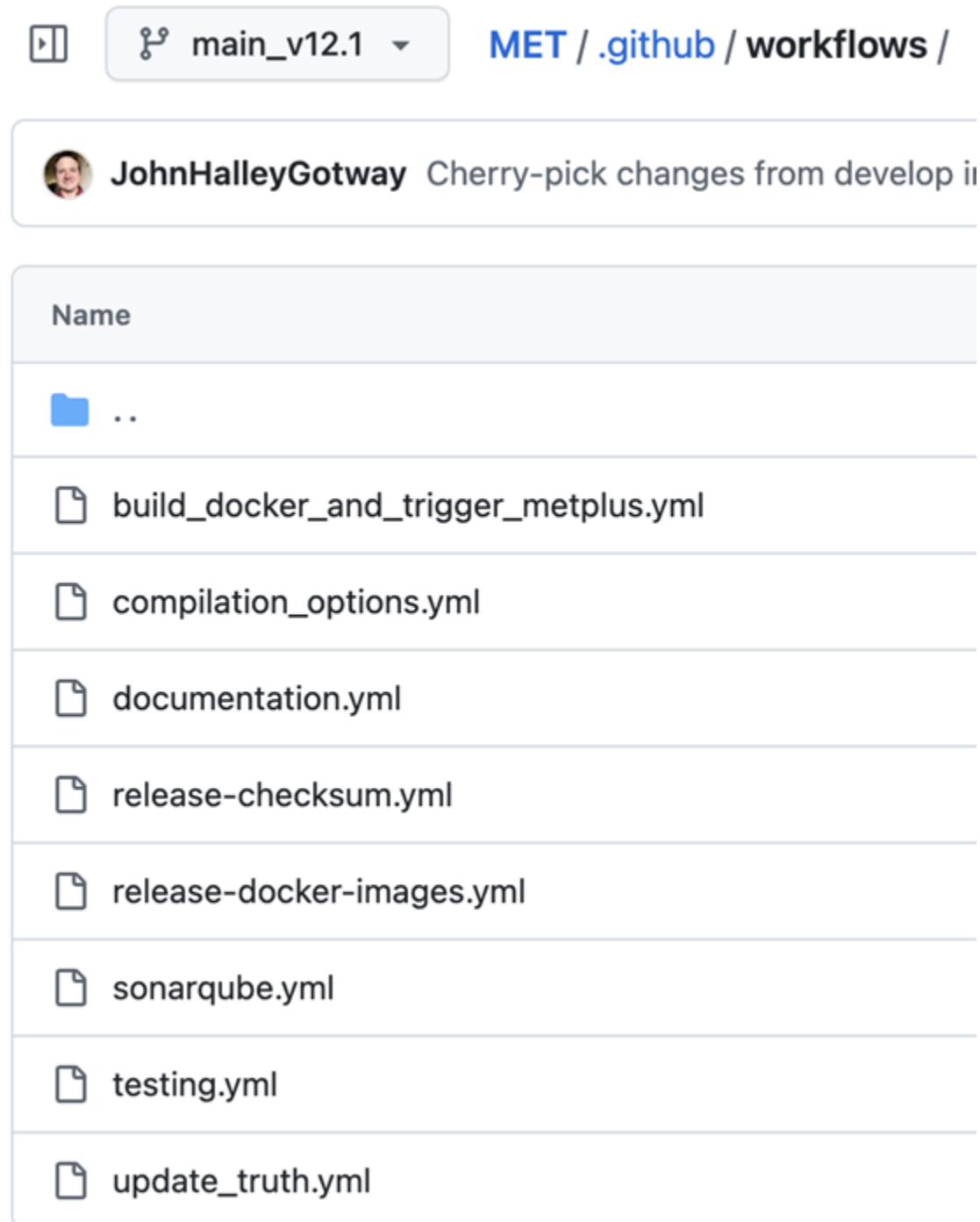
Contributor's Guide



GitHub Actions Overview

workflow automation

- METplus (9)
- MET (8) 
- METexpress (9)
- METviewer (7)
- METplotpy (6)
- METcalcpy (8)
- METdataio (9)
- METbaseimage (3)



The screenshot shows a GitHub repository interface. At the top, there's a breadcrumb trail: `main_v12.1` (selected) > `MET` / `.github` / `workflows` / `main_v12.1`. Below this, a commit by **JohnHalleyGotway** is shown with the message "Cherry-pick changes from develop into main_v12.1". The main content area displays a list of workflow files under the heading "Name". The files listed are:

- `..` (directory)
- `build_docker_and_trigger_metplus.yml`
- `compilation_options.yml`
- `documentation.yml`
- `release-checksum.yml`
- `release-docker-images.yml`
- `sonarqube.yml`
- `testing.yml`
- `update_truth.yml`

GHA: Release Checksums

add md5 checksums

Jul 30
georgemccabe
<> v12.1.0
5bbd804
Commits
Compare

MET-12.1.0

Please see the [Release Notes](#), [Development Issues](#), and User's Guide ([html](#), [pdf](#)).

Assets 4

checksum_tar.txt	84 Bytes	Jul 30
checksum_zip.txt	81 Bytes	Jul 30
Source code (zip)		Jul 30
Source code (tar.gz)		Jul 30

```
40
41 # create checksum from both files
42 echo Creating $cs_zip_filename with checksums for ZIP file
43 sha256sum ${dl_filename}.zip > $cs_zip_filename
44
45 echo Creating $cs_tar_filename with checksums for TAR file
46 sha256sum ${dl_filename}.tar.gz > $cs_tar_filename
47
```

release-checksum.yml

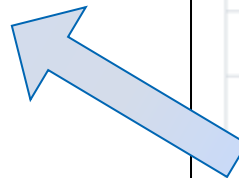
```
1 name: Add checksum to release
2
3 on:
4   release:
5     types:
6       - published
7
8 jobs:
9   add-checksum:
10    runs-on: ubuntu-latest
11    steps:
12      - uses: dtcenter/metplus-action-release-checksum@v2
13        with:
14          token: ${ secrets.METPLUS_BOT_TOKEN }
```

metplus-action-release-checksum Public Edit Pins Watch 1

main Go to file + Code

georgemccabe update version history for v2 9dd7dc5 · last year

Dockerfile	install curl	3 years ago
README.md	update version history for v2	last year
action.yml	test to check env var values are corre...	3 years ago
entrypoint.sh	removed GITHUB_OAUTH_BASIC wh...	last year



GHA: Documentation

check for warnings

The screenshot shows the GitHub Actions interface. The top navigation bar includes 'Code', 'Issues' (327), 'Pull requests', 'Actions', 'Projects' (11), and 'Wiki'. The 'Actions' tab is active, displaying a list of workflows on the left: 'All workflows', 'Add checksum to release', 'Build Docker Image and Trigger METplus Workflow', 'Compilation Options', 'Create Release Docker Images', 'Documentation' (selected), 'SonarQube Scan', 'Testing', and 'Update Truth Data'. The main area shows the 'Documentation' workflow with a search bar and '887 workflow runs'. A message states: 'This workflow has a workflow_dispatch event trigger.' A 'Run workflow' button is visible. A modal window titled 'Use workflow from' is open, showing 'Branch: main_v12.2' and a 'Run workflow' button. Below the modal, a workflow run is listed: 'Merge branch' with a green checkmark, 'Documentation #3638: Commit 6d55cda pushed by georgemccabe', and a timestamp '13 AM'.

- Documentation
Documentation #3639: Manually run by JohnHalleyGotway main_v12.2
- Documentation
Documentation #3639: Manually run by JohnHalleyGotway main_v12.2

documentation.yml

```
1  name: Documentation
2  on:
3    push:
4      branches:
5        - develop
6        - develop-ref
7        - feature_*
8        - main_*
9        - bugfix_*
10     paths:
11       - docs/**
12     pull_request:
13       types: [opened, reopened, synchronize]
14     workflow_dispatch:
15
```

Artifacts

Produced during runtime

Name

Size

Digest

 documentation

6.59 MB

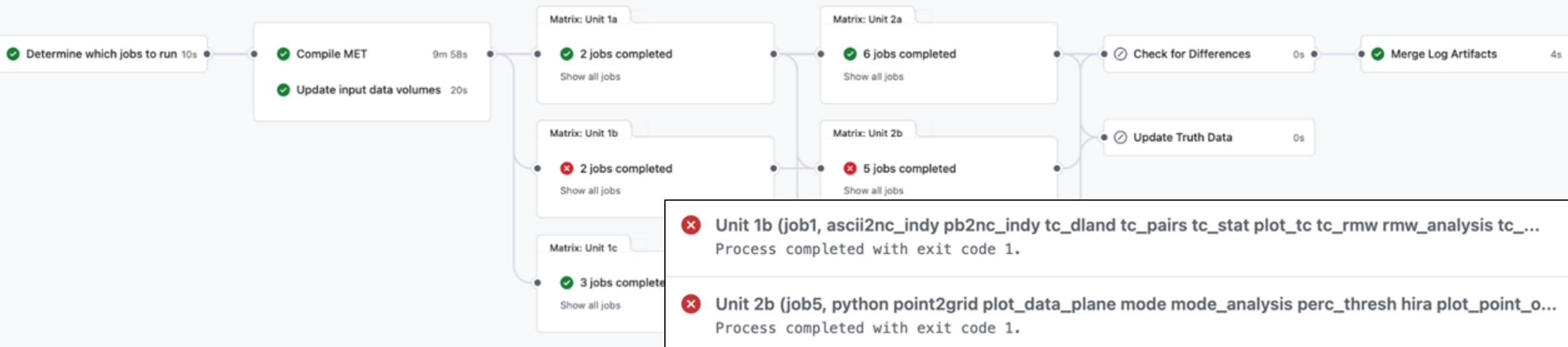
sha256:e517d00fbc35c88e84...



GHA: Testing Workflow

regression testing

testing.yml on: push	Code Compiles
testing.yml on: push	Tests Run, No Diffs
testing.yml on: workflow_dispatch	Tests Run, Diffs Found
testing.yml on: workflow_dispatch	Tests Failed



GHA: Trigger METplus

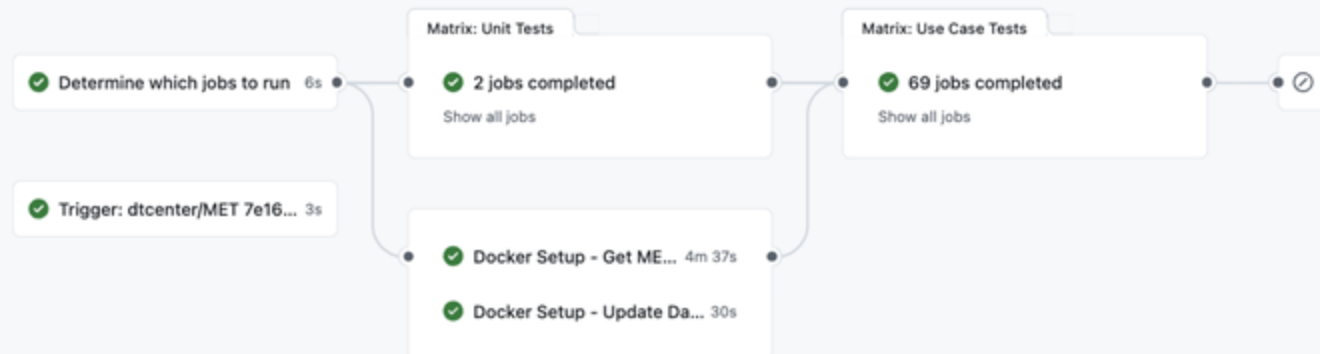
integration testing

```
# trigger workflow using gh cli
json_inputs="{\"actor\": \"${GITHUB_ACTOR}\", \"ref\": \"${GITHUB_REF}\", \"re
echo \"Triggering METplus testing workflow on ${metplus_branch} branch using
echo \"${json_inputs}\" | gh workflow run testing.yml --repo dtcenter/METplus
```

Manually triggered last week	Status	Total duration	Artifacts
 metplus-bot  89d9a66 develop	Success	1h 23m 24s	69

testing.yml

on: workflow_dispatch



build_docker_and_trigger_metplus.yml

```
1   name: Build Docker Image and Trigger METplus Workflow
2
3   on:
4     push:
5       branches:
6         - develop
7         - 'main_v[0-9]+.[0-9]+'
8       paths-ignore:
9         - 'docs/**'
10
11    workflow_dispatch:
12
13    env:
14      DOCKERHUB_REPO: dtcenter/met-dev
15
16    jobs:
17
18    build_met_docker:
19
20    trigger_metplus:
21      name: Trigger METplus testing workflow
22      runs-on: ubuntu-latest
23      needs: build_met_docker
24      steps:
25        - uses: dtcenter/metplus-action-trigger-use-cases@v1
26          with:
27            token: ${ secrets.METPLUS_BOT_TOKEN }
```

GHA: Release Docker Images

rebuild images weekly

```
1  name: Create Release Docker Images
2
3  on:
4
5      push:
6          tags:
7              - v*
8
9      workflow_dispatch:
10         inputs:
11             release_version:
12                 description: 'Comma-separated list of versions to build, e.g. v3.3,v3.4,v3.5'
13                 default: 'v3.3,v3.4,v3.5'
14                 required: true
15                 type: string
16             update_latest:
17                 description: 'Update "X.Y-latest" images.'
18                 default: false
19                 required: true
20                 type: boolean
21
22     schedule:
23         - cron: "30 00 * * 0"
```

Rebuild Supported
Versions Weekly

METbaseimage @
00:30

MET + METviewer @
03:30

METplus @ 06:30



dtcenter/metplus Sponsored OSS

By [NCAR/NOAA](#) · Updated 2 days ago

Python scripting infrastructure for MET tools. Use tags named X.Y-latest or X.Y.Z.

[IMAGE](#)

[MACHINE LEARNING & AI](#) [DATA SCIENCE](#) [MONITORING & OBSERVABILITY](#)

☆2 ↓ 10K+

TAG

6.1.0

Last pushed 2 days by [dtcenter](#)

docker pull dtcenter/metplus:6.1.0

Digest	OS/ARCH	Vulnerabilities	Last pull	Compressed size
e73d4aa28b3c	linux/amd64	0 1 1 186 2	2 days	2 GB

5.1.0

Last pushed over 2 years by [dtcenter](#)

docker pull dtcenter/metplus:5.1.0

Digest	OS/ARCH	Vulnerabilities	Last pull	Compressed size
302823f764cb	linux/amd64	0 3 2 5 6	about 2 months	1.39 GB



GHA: CVE Scanning

vulnerabilities

Jobs **METviewer**

- ✓ Define Matrix
- ✓ Build, Scan, and Push Images (v6.0.3)
- ✗ Build, Scan, and Push Images (v6.1.0)

Configuration:

- Fail on Critical CVEs: true
- Fail on High CVEs: false

Scanning image dtcenter/metviewer:6.0.3

Logging to /home/runner/work/METviewer/security-scan-logs/CVE_Scan_dtcenter_metviewer_6.0.3.log

Found 899 CVEs for image dtcenter/metviewer:6.0.3: 0 Critical, 41 High, 82 Medium, 34 Low, 742 Negligible

Warning: Found 41 High CVEs for image dtcenter/metviewer:6.0.3

Found 964 CVEs for image dtcenter/metviewer:6.1.0: 5 Critical, 59 High, 93 Medium, 37 Low, 770 Negligible

Error: Found 5 Critical CVEs for image dtcenter/metviewer:6.1.0

release-docker-images.yml (development-docker-images.yml)

```
- name: Scan Docker Images
  id: security-scan
  continue-on-error: false
  uses: dtcenter/metplus-action-scan-docker-images@v1
  with:
    images: |
      [
        "${{ env.DOCKERHUB_BASE_REPO }}:${{ env.VERSION_NO_V }}",
        "${{ env.DOCKERHUB_ANALYSIS_REPO }}:${{ env.VERSION_NO_V }}"
      ]
    fail-on-critical: "true"
    log-artifact-name: 'security-scan-logs_${{ env.VERSION_NO_V }}'
```



```
if ! grype $image > "$log_file" 2>&1; then
```

Thank You

Docs | <https://metplus.readthedocs.io>

Web | <https://dtcenter.org/community-code/metplus>

Help | <https://github.com/dtcenter/METplus/discussions>